Chapter 1

Game Design Document

George Larkwright

Last Updated: 11/07/2019

Game Overview

Chapter 1 is an educational dungeon crawler, in which players guide a self-designed adventurer through different worlds, encountering enemies, puzzles and other obstacles. Rather than using combat, items or other RPG staples to pass obstacles, players must write out their own solutions using nothing more than a keyboard and their imagination.

The game uses natural language processing to generate new writing prompts based on what the player has already written. As players progress through each dungeon they build up a story that they can print out at the end of the game.

Design Values

Before beginning work on the game I settled on the following design values. These will guide what I was trying to create in this project.

Approachable

First and foremost I want the game to be attractive to young people; despite being an educational game, this should be something that they want to play. I imagine a lot of this will be achieved through aesthetics – cartoonish visuals, satisfying effects, and an exciting world to explore. The core game loop should also be fun to play, and encourage repeat playthroughs.

Accessible

I want to create a game that is easy for young people to play, even if they might not be the top of their class in Literacy. The whole point of learning games is just that – they are about learning. The game should not be limited to those students who have learned a lot already.

Structured Creativity

I did a great deal of research into what creativity is and how we can encourage it, especially through education, before beginning work on this project. I ran multiple workshops with young people aged 8-11, and wrote these up into a case study. I also used knowledge gained from two years in the teaching profession.

What I discovered was that young people respond well to freedom to express themselves. However, complete freedom means that learning objectives are not met. This is why the concept of 'structured creativity' is so important to this project – I want to give players freedom to express themselves through writing, but build a structure to this expression so that they have a tangible end product at the end of the game.

Encouraging and Inspiring

I want the game to inspire players to create their own stories outside of the game, after playing. After completing the game players will be presented with their complete story, and will have the option to print this out. They will be encouraged to share this story with others, and also given tips on how to improve. Positive reinforcement is really important when encouraging young people to persevere with something they might find a challenge.

Impactful learning experience

As an education game first and foremost, this project should provide an impactful learning experience for players. Alongside giving players the kind of scaffolding to start and continue writing their story (through the generated prompts), the game should offer guidance on how players can improve their writing.

Encourage adult involvement

The problem with a lot of learning games is that they are seen as an activity for young people to do on their own – a way of learning without any adult input. I don't believe this is an effective way of supporting a young person to learn. I want this game to encourage adults to get involved in the learning process, supporting the young person as they write their stories, and then reading the finish product afterwards as well.

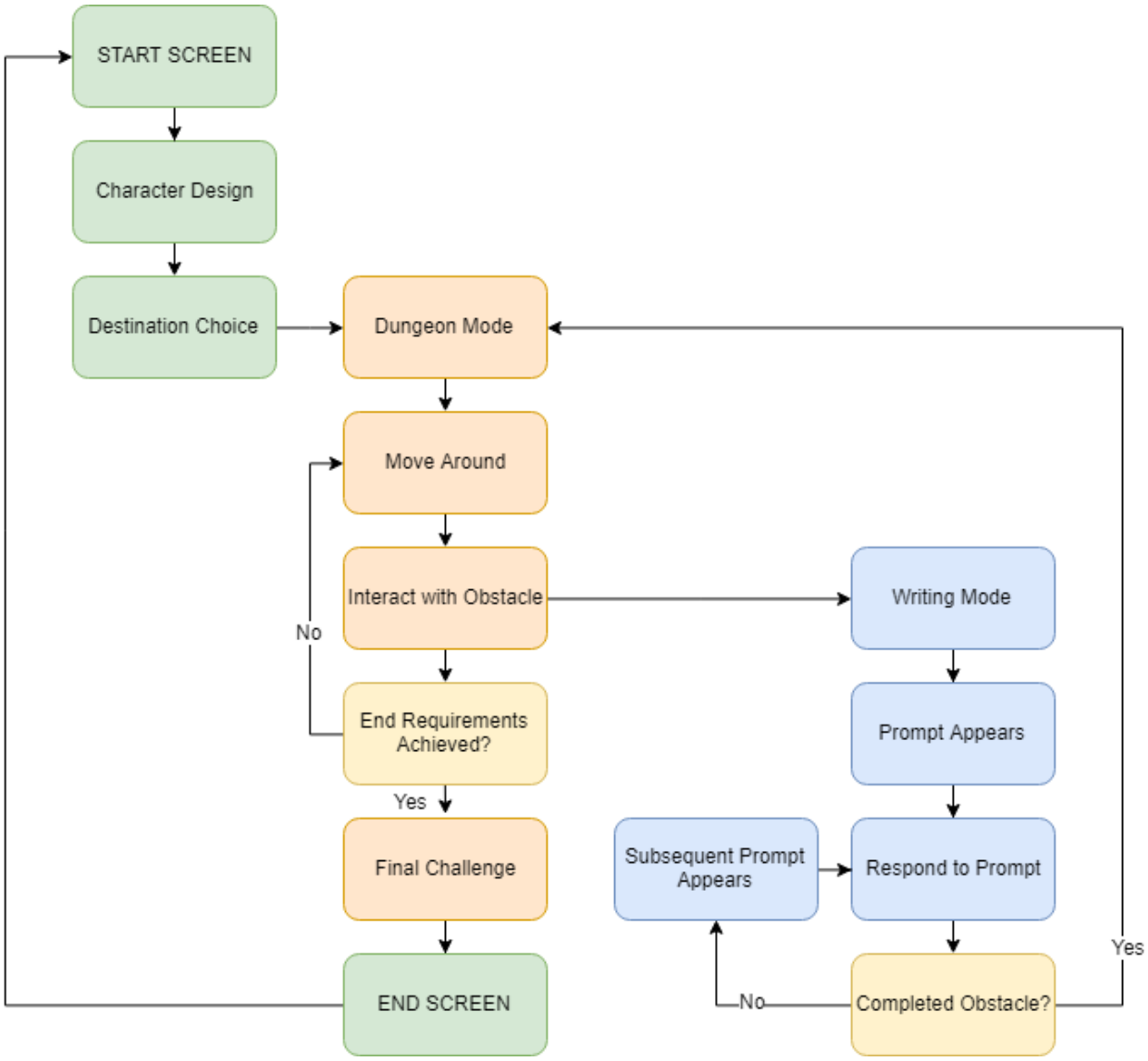
Gameplay Flow

The game will have a very simple flow, as I do not want this to complicate things for the player; the focus should be on the writing sections.

The game will have two modes: Dungeon (or Adventure) Mode, and Writing Mode.

In Dungeon Mode, the player explores an overworld full of different characters, obstacles, and other interactable objects.

Upon interacting with an object, the game switches to Writing Mode, and the player must write answers to prompts to get past the object.



1. **Character creation**: players must fill out a character sheet and choose a look for their hero.

2. **Destination choice**: players can choose which destination they want to go to. For the early build, there will only be one destination available.

3. **Dungeon Crawling begins! Tutorial onboarding.** Player's companion introduces them to walking around the overworld, and interacting with objects.

4. **Move Around**: players use the arrow keys to move around the overworld, and can interact with objects with space.

5. **Interact with object**: When a player interacts with an object, they move to Writing Mode.

6. **Prompt appears**: The player must complete a writing prompt to 'defeat' an obstacle.

7. The game checks to see if the response to the prompt is adequate to pass the obstacle. If not, a subsequent prompt appears. The player continues to respond to prompts, building up a paragraph response that will fit into their story.

8. Once the obstacle is completed, players return to the **Dungeon Mode** overworld, where they can continue exploring and looping through other obstacles.

9. If all the obstacles have been interacted with, then the players complete a **Final Challenge** (in writing mode). This ends the story

10. **End Screen**: the player is faced with the end screen, on which they can read their completed story (compiled from their writing prompts) and choose to print it before returning to the start screen.

Gameplay: Dungeon/Adventure Mode

The Dungeon Crawling Mode (I'm thinking of labelling this 'Adventure Mode' to be more appropriate for young people) is the game's 'overworld'. Players explore the overworld and the things they encounter form the building blocks for their story. Players not only get to write the story, they get to live the story. Each overworld should contain different objects and paths to make the world feel new each time.

The features of this mode will be quite limited as the mode's purpose is to facilitate the writing sections of the game.

Basic Elements

Map: The overworld will have an isometric tile-based map. The tile-based system means it is easy to design levels on a grid, blocking out different pathways for the players to go, and different obstacles they might face on a tile-by-tile basis. The bounds of the map will be marked by walls. The map should not be too big so as to lose a player's attention, but not too small to prevent the creation of a full story. The map will be made on the Tiled Map Editor.

Playable Character: The player will design their character at the beginning of the game, and this will then be translated into a sprite that they can guide around the overworld.

Obstacles: The player will encounter a variety of obstacles in the overworld that they will have to pass. Interacting with an obstacle will trigger the game to switch to writing mode, upon which the player is prompted to write how they will get past the obstacle.

NPCs: Some worlds may have NPC characters that the player can talk to. These would also prompt the player to write interactions, with prompts geared at conversation.

Inventory Items: I am undecided on including inventory items that can be references by the player in later written prompts. This might be a nice feature, but might complicate things; I want the player to use their writing to solve problems, rather than having an inventory item that does the same thing.

Dialogue Box: The bottom 25% of the screen will be taken up by a dialogue section, which will show dialogue whenever the player moves close to an object, giving added character to the world. The dialogue box will feature an animated avatar of the player character when they speak.

Scoring: I will not be using a score to incentivise playing this game; the incentive should be the desire to finish your story.

Controls

The overworld controls are simple. The playable character will move around the map by clicking the tile you wish to move to. When next to an interactable object, pressing the space bar will switch to the writing mode.

Gameplay: Writing Mode

The Writing Mode is the core of the game. Upon interacting with an object in the overworld, the player will be prompted to write a response to the object. For example, the object might be a ravine, and the player might have to describe how they get over the ravine. Upon completing a prompt, the player receives additional prompts to follow based on what they first wrote. They might be asked to describe the rope they used to swing across.

After completing a nominal number of prompts, the object will be classed as 'completed' and players will be able to return to the overworld mode to continue their exploration. The completed passage of writing will be added to their story.

Basic Elements

Prompts: Each object will have an initial prompt attached to it, which the player must answer. New prompts are then generated, which relate to what the player has already written.

Writing Window: This is the space in which the player's writing appears as they type. It will function like an ordinary text box, but be style to fit the game's aesthetic rather than a boring Word Doc. Players will be able to submit what they've written with a button click.

Support Prompts: After answering the initial prompts the player may receive addition ideas e.g. 'Try using a new adjective' to support their writing. Not yet clear how 'hands-on' this feature will be.

Imagery: The Writing Mode will also feature a section for showing an image of the object being interacted with, or the player avatar where appropriate.

Natural Language Processing

The writing mode will use natural language processing and parts-of-speech tagging to parse what the player write, and generates prompts to continue writing based on this.

Controls

The player will use the keyboard to input text, and click a submit button to submit – easy!

Visual Aesthetic

Graphics

I want the game to be viable on school laptops rather than high-end machines, so the graphical style won't be too complex. The game will take place in a 4:3 ratio window.

The overworld will be made out of isometric tiles, with a low-detail pixelated look, similar to that used in *Into The Breach* (right).

The overworld will be occupied by different obstacles, all given a tile. They might be built into the tile, such as a crevasse or a mountain, or they might just be an object or NPC on-top of the tile. These will be pixel sprites, styled in 3D to fit the isometric POV.

Rather use 3D sprites, I will make the player character a 2D sprite so that they stand out from the backdrop. I think this will also make the movement feel a lot freer and less stuck to the tiles. The movement of old Lucas Arts adventure games is a good reference point (*Ben There Dan That* (below left) captures this depth within a 3D environment using the Adventure Game Studio toolset. More recently, *Broken Age* (below right) did a great job of using flat art but giving the world depth.

The game will include illustrated portraits of characters, including the player character. These will be used in dialogue sections.

Before beginning the game, players will be able to design their own characters in a Mii-like character designer. There will be a number of options for hair, eyes, mouth, etc, with the potential for about 12 different faces (this can easily be expanded on beyond the initial prototype). These portraits will have a hand-painted cartoon look, similar to *Animal Crossing* villagers (right).

UI

The UI will be flat and simple, so as not to detract from the game elements and allow the player to focus on the writing. For a colour scheme, I am keen on a palette of dark green and black/white, depending on user choice (dark mode if you will).

I have constructed two simple wireframe schematics for reference of the layout of each mode (colours not indicative of final UI colours).

Dungeon/Adventure Mode:

Writing Mode: